

Quantium Analysis 2026

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Summary: This is an inventory analysis conducted on the vendor's potato chip stock, evaluating the performance of the potato chip inventory over a year.

```
In [101... # Pandas and Numpy imported for analysis
import pandas as pd
import numpy as np
```

```
In [102... # Loaded needed dataframes for analysis
transactionData = pd.read_csv("C://Users//Nicho//Downloads//QuantiumAnalysis2026//QV
customerData = pd.read_csv("C://Users//Nicho//Downloads//QuantiumAnalysis2026//QVI_p
```

```
In [103... # Identify types of each column
## Note: Date needs to be altered to a date format
transactionData.dtypes
```

```
Out[103... DATE                int64
STORE_NBR                int64
LYLTY_CARD_NBR          int64
TXN_ID                  int64
PROD_NBR                int64
PROD_NAME               object
PROD_QTY               int64
TOT_SALES              float64
dtype: object
```

```
In [104... # check customer data as well
customerData.dtypes # no apparent issues
```

```
Out[104... LYLTY_CARD_NBR          int64
LIFESTAGE              object
PREMIUM_CUSTOMER       object
dtype: object
```

```
In [105... # explore the dataframe
transactionData.head(10) # date seems to be in a numeric format instead of a clear
```

Out[105...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_QTY
0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g	2
1	43599	1	1307	348	66	CCs Nacho Cheese 175g	3
2	43605	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	2
3	43329	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	5
4	43330	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	3
5	43604	4	4074	2982	57	Old El Paso Salsa Dip Tomato Mild 300g	1
6	43601	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	1
7	43601	4	4196	3539	24	Grain Waves Sweet Chilli 210g	1
8	43332	5	5026	4525	42	Doritos Corn Chip Mexican Jalapeno 150g	1
9	43330	7	7150	6900	52	Grain Waves Sour Cream&Chives 210G	2

In [106...

```
# exploring the date column to understand format
earliestdate = transactionData['DATE'].min()
latestdate = transactionData['DATE'].max()
print(f" Min: {earliestdate} \n Max: {latestdate} r") # 364 date might indicate a y
```

Min: 43282

Max: 43646 r

In [107...

```
# identifying unique values for each column:
# Note: Date has 364 unique date Numbers and the max-min difference is 364, indicat
transactionData.nunique() # further research indicates this format is likely Excel
```

```
Out[107...  DATE                364
          STORE_NBR      272
          LYLTY_CARD_NBR  72637
          TXN_ID         263127
          PROD_NBR        114
          PROD_NAME       114
          PROD_QTY         6
          TOT_SALES       112
          dtype: int64
```

```
In [108... # Creating a copy for version control
transactionData2 = transactionData.copy()
transactionData2['DATE'] = pd.to_datetime(transactionData2['DATE'], unit='D', origi
# Dates are to be interpreted as 43282 days passed since 1899-12-30 and so forth
transactionData2.head(10)
```

Out[108...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
5	2019-05-19	4	4074	2982	57	Old El Paso Salsa Dip Tomato Mild 300g	
6	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
7	2019-05-16	4	4196	3539	24	Grain Waves Sweet Chilli 210g	
8	2018-08-20	5	5026	4525	42	Doritos Corn Chip Mexican Jalapeno 150g	
9	2018-08-18	7	7150	6900	52	Grain Waves Sour Cream&Chives 210G	

In [109...

```
transactionData2.dtypes # Data is now a date format
```

Out[109...

```
DATE                datetime64[ns]
STORE_NBR           int64
LYLTY_CARD_NBR      int64
TXN_ID              int64
PROD_NBR            int64
PROD_NAME           object
PROD_QTY            int64
TOT_SALES           float64
dtype: object
```

```
In [110... # exploring the date column under the new format
# note: the diff is not inclusive of the start date 2018-07-01 and is truly represe
# this counts 2018-07-02 to 2019-06-30 (Fencepost Error)
earliestdate = transactionData2['DATE'].min()
latestdate = transactionData2['DATE'].max()
print(f" Min: {earliestdate} \n Max: {latestdate} \n diff: {latestdate - earliestda
# with this Fencepost Error noted, we can conclude that a day is missing from the d
```

Min: 2018-07-01 00:00:00

Max: 2019-06-30 00:00:00

diff: 364 days 00:00:00

unique: 364

```
In [111... transactionData2.isnull().sum()
```

```
Out[111... DATE                0
STORE_NBR             0
LYLTY_CARD_NBR        0
TXN_ID                0
PROD_NBR              0
PROD_NAME             0
PROD_QTY              0
TOT_SALES             0
dtype: int64
```

```
In [112... # Identifying the missing value
accurate_range = pd.date_range(start='2018-07-01', end='2019-06-30')
missing_date = accurate_range.difference(transactionData2['DATE'])
print(f"the missing date is: {missing_date}")
```

the missing date is: DatetimeIndex(['2018-12-25'], dtype='datetime64[ns]', freq='D')

```
In [113... transactionData2.head()
```

```
Out[113...   DATE  STORE_NBR  LYLTY_CARD_NBR  TXN_ID  PROD_NBR  PROD_NAME  PROD_
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-14	1	1307	348	66	CCs Nacho Cheese 175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
4	2018-08-18	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	

```
In [114... # Ensure store numbers are representative of actual locations
print(f"Stores: {transactionData2['STORE_NBR'].nunique()} \nMaxStoreNumber: {transa
# There are 272 unique stores, and store numbers range from 1 to 272, indicating pr
```

```
Stores: 272
MaxStoreNumber: 272
MinStoreNumber: 1
```

```
In [115... # Produce QTY verification: numbers should be at the lowest 1
print(f"MinProductQuantity: {transactionData2['PROD_QTY'].min()} \nMaxProductQuanti
```

```
MinProductQuantity: 1
MaxProductQuantity: 200
```

```
In [116... # Examining product names to ensure they are correct
transactionData2['PROD_NAME'].describe()
# we are looking at 114 different products
```

```
Out[116... count                264836
unique                  114
top      Kettle Mozzarella  Basil & Pesto 175g
freq                3304
Name: PROD_NAME, dtype: object
```

```
In [117... transactionData2['PROD_NAME'].value_counts()
# woolworths medium Salsa, indicates non-chip products may be included
```

```

Out[117...  PROD_NAME
Kettle Mozzarella Basil & Pesto 175g 3304
Kettle Tortilla ChpsHny&Jlpno Chili 150g 3296
Cobs Popd Swt/Chlli &Sr/Cream Chips 110g 3269
Tyrrells Crisps Ched & Chives 165g 3268
Cobs Popd Sea Salt Chips 110g 3265
Kettle 135g Swt Pot Sea Salt 3257
Tostitos Splash Of Lime 175g 3252
Infuzions Thai SweetChili PotatoMix 110g 3242
Smiths Crnkle Chip Orgnl Big Bag 380g 3233
Thins Potato Chips Hot & Spicy 175g 3229
Kettle Sensations Camembert & Fig 150g 3219
Doritos Corn Chips Cheese Supreme 170g 3217
Pringles Barbeque 134g 3210
Doritos Corn Chip Mexican Jalapeno 150g 3204
Kettle Sweet Chilli And Sour Cream 175g 3200
Smiths Crinkle Chips Salt & Vinegar 330g 3197
Thins Chips Light& Tangy 175g 3188
Dorito Corn Chp Supreme 380g 3185
Pringles Sweet&Spcy BBQ 134g 3177
Infuzions BBQ Rib Prawn Crackers 110g 3174
Tyrrells Crisps Lightly Salted 165g 3174
Kettle Sea Salt And Vinegar 175g 3173
Doritos Corn Chip Southern Chicken 150g 3172
Twisties Chicken270g 3170
Twisties Cheese Burger 250g 3169
Grain Waves Sweet Chilli 210g 3167
Pringles SourCream Onion 134g 3162
Doritos Corn Chips Nacho Cheese 170g 3160
Cobs Popd Sour Crm &Chives Chips 110g 3159
Kettle Original 175g 3159
Pringles Original Crisps 134g 3157
Cheezels Cheese 330g 3149
Kettle Honey Soy Chicken 175g 3148
Kettle Tortilla ChpsBtroot&Ricotta 150g 3146
Tostitos Smoked Chipotle 175g 3145
Infzns Crn Crnchers Tangy Gcamole 110g 3144
Smiths Crinkle Original 330g 3142
Kettle Tortilla ChpsFeta&Garlic 150g 3138
Infuzions SourCream&Herbs Veg Strws 110g 3134
Kettle Sensations Siracha Lime 150g 3127
Old El Paso Salsa Dip Chnky Tom Ht300g 3125
Doritos Corn Chips Original 170g 3121
Doritos Mexicana 170g 3115
Twisties Cheese 270g 3115
Thins Chips Seasonedchicken 175g 3114
Old El Paso Salsa Dip Tomato Med 300g 3114
Pringles Mystery Flavour 134g 3114
Grain Waves Sour Cream&Chives 210G 3105
Pringles Chicken Salt Crips 134g 3104
Thins Chips Salt & Vinegar 175g 3103
Pringles Slt Vingar 134g 3095
Old El Paso Salsa Dip Tomato Mild 300g 3085
Kettle Sensations BBQ&Maple 150g 3083
Pringles Sthrn FriedChicken 134g 3083
Tostitos Lightly Salted 175g 3074

```

Doritos Cheese	Supreme 330g	3052
Kettle Chilli 175g		3038
Smiths Chip Thinly	Cut Original 175g	1614
Snbts Whlgrn Crisps	Cheddr&Mstrd 90g	1576
Natural Chip Co	Tmato Hrb&Spce 175g	1572
Burger Rings 220g		1564
Natural ChipCo Sea	Salt & Vinegr 175g	1550
CCs Tasty Cheese	175g	1539
RRD SR Slow Rst	Pork Belly 150g	1526
Smiths Thinly Cut	Roast Chicken 175g	1519
RRD Sweet Chilli &	Sour Cream 165g	1516
Woolworths Cheese	Rings 190g	1516
CCs Original 175g		1514
RRD Honey Soy	Chicken 165g	1513
Smith Crinkle Cut	Mac N Cheese 150g	1512
WW Supreme Cheese	Corn Chips 200g	1509
Infuzions Mango	Chutny Papadums 70g	1507
RRD Chilli&	Coconut 150g	1506
Smiths Crinkle Cut	Snag&Sauce 150g	1503
Red Rock Deli Sp	Salt & Truffle 150G	1498
CCs Nacho Cheese	175g	1498
WW Original Corn	Chips 200g	1495
Red Rock Deli Thai	Chilli&Lime 150g	1495
Woolworths Mild	Salsa 300g	1491
Smiths Crinkle Cut	Chips Barbecue 170g	1489
WW Original Stacked	Chips 160g	1487
Smiths Crinkle Cut	Chips Chicken 170g	1484
WW Sour Cream &Onion	Stacked Chips 160g	1483
Smiths Crinkle Cut	Chips Chs&Onion170g	1481
Cheetos Chs & Bacon	Balls 190g	1479
RRD Salt & Vinegar	165g	1474
RRD Lime & Pepper	165g	1473
Smiths Chip Thinly	S/Cream&Onion 175g	1473
Doritos Salsa Mild	300g	1472
Smiths Crinkle Cut	Tomato Salsa 150g	1470
WW D/Style Chip	Sea Salt 200g	1469
Natural Chip	Compny SeaSalt175g	1468
GrnWves Plus Btroot	& Chilli Jam 180g	1468
WW Crinkle Cut	Chicken 175g	1467
Smiths Thinly	Swt Chli&S/Cream175G	1461
Smiths Crinkle Cut	Chips Original 170g	1461
Natural ChipCo	Hony Soy Chckn175g	1460
Red Rock Deli SR	Salsa & Mzzrlla 150g	1458
Smiths Crinkle Cut	Salt & Vinegar 170g	1455
RRD Steak &	Chimuchurri 150g	1455
Cheezels Cheese Box	125g	1454
Smith Crinkle Cut	Bolognese 150g	1451
Doritos Salsa	Medium 300g	1449
Cheetos Puffs 165g		1448
Thins Chips	Originl saltd 175g	1441
Smiths Chip Thinly	CutSalt/Vinegr175g	1440
Smiths Crinkle Cut	French OnionDip 150g	1438
Red Rock Deli Chikn&	Garlic Aioli 150g	1434
Sunbites Whlegrn	Crisps Frch/Onin 90g	1432
RRD Pc Sea Salt	165g	1431
Woolworths Medium	Salsa 300g	1430


```
NCC Sour Cream & Garden Chives 175g    1419
French Fries Potato Chips 175g          1418
WW Crinkle Cut Original 175g           1410
Name: count, dtype: int64
```

```
In [118... # Ensuring all products are chips
# Split all unique product names into individual words
product_words = (
    transactionData2['PROD_NAME']
    .unique()                # get unique product names
    .tolist()               # convert to list
)

# Join all product names into one string, then split into individual words
product_words = pd.Series(
    ' '.join(product_words).split()
)

print(product_words.value_counts())
```

175g	26
Chips	21
150g	19
&	17
Smiths	16
Crinkle	14
Cut	14
Kettle	13
Salt	12
Cheese	12
Original	10
Salsa	9
Chip	9
Doritos	9
Pringles	8
134g	8
170g	8
Corn	8
165g	8
RRD	8
WW	7
Chicken	7
110g	7
300g	6
Sea	6
Sour	6
Thins	5
Vinegar	5
Chilli	5
Thinly	5
Crisps	5
Cream	4
Rock	4
Deli	4
Supreme	4
Infuzions	4
Natural	4
Red	4
330g	4
Dip	3
Old	3
Tomato	3
200g	3
Mild	3
El	3
CCs	3
Sweet	3
Twisties	3
Lime	3
Cobs	3
Sensations	3
Popd	3
Tostitos	3
Soy	3
Paso	3
Woolworths	3

Tortilla	3
ChipCo	2
And	2
Thai	2
Tyrrells	2
Lightly	2
160g	2
Swt	2
BBQ	2
SR	2
Honey	2
French	2
Cheezels	2
Salted	2
Smith	2
Nacho	2
Waves	2
Potato	2
Cheetos	2
Grain	2
380g	2
Tangy	2
90g	2
Burger	2
Rings	2
190g	2
Chives	2
Medium	2
Barbecue	1
Crips	1
Stacked	1
CutSalt/Vinegr175g	1
Aioli	1
D/Style	1
BBQ&Maple	1
Fries	1
Med	1
Pc	1
Chilli&	1
Coconut	1
Hot	1
Spicy	1
Crm	1
Belly	1
Puffs	1
&Chives	1
Crnkle	1
Orgnl	1
Big	1
Bag	1
Pork	1
Ched	1
Splash	1
Chimuchurri	1
Strws	1
ChpsFeta&Garlic	1

Mango	1
Chutny	1
Papadums	1
70g	1
Steak	1
Sunbites	1
Hrb&Spce	1
Whlegrn	1
Frch/Onin	1
Snag&Sauce	1
&OnionStacked	1
Pepper	1
Vinegr	1
Chikn&Garlic	1
Veg	1
SourCream&Herbs	1
Of	1
Snbts	1
ChpsBtroot&Ricotta	1
Tasty	1
Smoked	1
Chipotle	1
Barbeque	1
Mystery	1
Flavour	1
Whlgrn	1
Slow	1
Cheddr&Mstrd	1
Chs	1
Bacon	1
Balls	1
Rst	1
Slt	1
Vingar	1
Chs&Onion170g	1
SweetChili	1
Tmato	1
Co	1
125g	1
Infzns	1
Crn	1
Crnchers	1
Gcamole	1
Chilli&Lime	1
Sthrn	1
FriedChicken	1
Sweet&Spcy	1
Mzzrlla	1
Originl	1
saltd	1
Sp	1
Truffle	1
150G	1
Box	1
Southern	1
Garden	1

Mexican	1
Compny	1
SeaSalt175g	1
S/Cream&Onion	1
ChpsHny&Jlpno	1
Chili	1
210g	1
Jalapeno	1
NCC	1
Cream&Chives	1
210G	1
Siracha	1
270g	1
Light&	1
220g	1
Chli&S/Cream175G	1
Mexicana	1
OnionDip	1
SourCream	1
Plus	1
Btroot	1
Jam	1
180g	1
135g	1
Pot	1
Onion	1
Crackers	1
250g	1
Chnky	1
Tom	1
Ht300g	1
Swt/Chlli	1
&Sr/Cream	1
GrnWves	1
Prawn	1
Hony	1
Basil	1
Chckn175g	1
Dorito	1
Chp	1
Chicken270g	1
Roast	1
Mozzarella	1
Pesto	1
Rib	1
PotatoMix	1
Camembert	1
Fig	1
Mac	1
N	1
Seasonedchicken	1
Bolognese	1

Name: count, dtype: int64

In [119... `import re`

```
# Remove digits
product_words = product_words[~product_words.str.contains(r'\d', regex=True)]

# Remove special characters (keep only alphabetic words)
product_words = product_words[product_words.str.match(r'^[a-zA-Z]+$')]

# View word frequency high to low
word_freq = product_words.value_counts().reset_index()
word_freq.columns = ['WORD', 'COUNT']
print(word_freq)
```

	WORD	COUNT
0	Chips	21
1	Smiths	16
2	Crinkle	14
3	Cut	14
4	Kettle	13
5	Salt	12
6	Cheese	12
7	Original	10
8	Doritos	9
9	Chip	9
10	Salsa	9
11	Corn	8
12	Pringles	8
13	RRD	8
14	Chicken	7
15	WW	7
16	Sea	6
17	Sour	6
18	Thinly	5
19	Crisps	5
20	Vinegar	5
21	Thins	5
22	Chilli	5
23	Supreme	4
24	Deli	4
25	Rock	4
26	Red	4
27	Cream	4
28	Natural	4
29	Infuzions	4
30	Mild	3
31	Old	3
32	Dip	3
33	Tostitos	3
34	Tomato	3
35	CCs	3
36	Paso	3
37	El	3
38	Tortilla	3
39	Soy	3
40	Cobs	3
41	Popd	3
42	Twisties	3
43	Lime	3
44	Woolworths	3
45	Sensations	3
46	Sweet	3
47	Swt	2
48	French	2
49	ChipCo	2
50	Tyrrells	2
51	Nacho	2
52	Smith	2
53	Honey	2
54	Cheetos	2

55	BBQ	2
56	Lightly	2
57	Salted	2
58	Medium	2
59	SR	2
60	Potato	2
61	Cheezels	2
62	Waves	2
63	Burger	2
64	Tangy	2
65	Chives	2
66	Grain	2
67	And	2
68	Thai	2
69	Rings	2
70	Barbecue	1
71	Rst	1
72	Belly	1
73	Puffs	1
74	Stacked	1
75	Crips	1
76	Bag	1
77	Pork	1
78	Splash	1
79	Chimuchurri	1
80	Of	1
81	Orgnl	1
82	Crnkle	1
83	Crm	1
84	Spicy	1
85	Pc	1
86	Hot	1
87	Coconut	1
88	Compny	1
89	Big	1
90	Slow	1
91	Tasty	1
92	Pepper	1
93	Sunbites	1
94	Papadums	1
95	Chutny	1
96	Mango	1
97	Strws	1
98	Veg	1
99	Vingar	1
100	Whlegrn	1
101	Slt	1
102	Med	1
103	Balls	1
104	Bacon	1
105	Steak	1
106	Chs	1
107	Vinegr	1
108	Whlgrn	1
109	Snbts	1
110	Ched	1

111	Flavour	1
112	Aioli	1
113	Mystery	1
114	Barbeque	1
115	Chipotle	1
116	Smoked	1
117	Mexican	1
118	Btroot	1
119	Fries	1
120	Roast	1
121	Dorito	1
122	Hony	1
123	OnionDip	1
124	Mexicana	1
125	Truffle	1
126	Sp	1
127	saltd	1
128	Originl	1
129	Mzzrlla	1
130	FriedChicken	1
131	Sthrn	1
132	Chili	1
133	Gcamole	1
134	Crnchers	1
135	Crn	1
136	Infzns	1
137	Box	1
138	Southern	1
139	Garden	1
140	NCC	1
141	Siracha	1
142	Chp	1
143	Mozzarella	1
144	Tmato	1
145	Basil	1
146	Co	1
147	Tom	1
148	Chnky	1
149	Onion	1
150	SourCream	1
151	Pot	1
152	Jam	1
153	Jalapeno	1
154	Plus	1
155	GrnWves	1
156	Crackers	1
157	Prawn	1
158	Rib	1
159	Seasonedchicken	1
160	N	1
161	Mac	1
162	Fig	1
163	Camembert	1
164	PotatoMix	1
165	SweetChili	1
166	Pesto	1

167 Bolognese 1

In [120... pd.set_option('display.max_rows', None)
word_freq

Out[120...

	WORD	COUNT
0	Chips	21
1	Smiths	16
2	Crinkle	14
3	Cut	14
4	Kettle	13
5	Salt	12
6	Cheese	12
7	Original	10
8	Doritos	9
9	Chip	9
10	Salsa	9
11	Corn	8
12	Pringles	8
13	RRD	8
14	Chicken	7
15	WW	7
16	Sea	6
17	Sour	6
18	Thinly	5
19	Crisps	5
20	Vinegar	5
21	Thins	5
22	Chilli	5
23	Supreme	4
24	Deli	4
25	Rock	4
26	Red	4
27	Cream	4
28	Natural	4
29	Infuzions	4

	WORD	COUNT
30	Mild	3
31	Old	3
32	Dip	3
33	Tostitos	3
34	Tomato	3
35	CCs	3
36	Paso	3
37	El	3
38	Tortilla	3
39	Soy	3
40	Cobs	3
41	Popd	3
42	Twisties	3
43	Lime	3
44	Woolworths	3
45	Sensations	3
46	Sweet	3
47	Swt	2
48	French	2
49	ChipCo	2
50	Tyrrells	2
51	Nacho	2
52	Smith	2
53	Honey	2
54	Cheetos	2
55	BBQ	2
56	Lightly	2
57	Salted	2
58	Medium	2
59	SR	2

	WORD	COUNT
60	Potato	2
61	Cheezels	2
62	Waves	2
63	Burger	2
64	Tangy	2
65	Chives	2
66	Grain	2
67	And	2
68	Thai	2
69	Rings	2
70	Barbecue	1
71	Rst	1
72	Belly	1
73	Puffs	1
74	Stacked	1
75	Crips	1
76	Bag	1
77	Pork	1
78	Splash	1
79	Chimuchurri	1
80	Of	1
81	Orgnl	1
82	Crnkle	1
83	Crm	1
84	Spicy	1
85	Pc	1
86	Hot	1
87	Coconut	1
88	Compny	1
89	Big	1

	WORD	COUNT
90	Slow	1
91	Tasty	1
92	Pepper	1
93	Sunbites	1
94	Papadums	1
95	Chutny	1
96	Mango	1
97	Strws	1
98	Veg	1
99	Vingar	1
100	Whlegrn	1
101	Slt	1
102	Med	1
103	Balls	1
104	Bacon	1
105	Steak	1
106	Chs	1
107	Vinegr	1
108	Whlgrn	1
109	Snbts	1
110	Ched	1
111	Flavour	1
112	Aioli	1
113	Mystery	1
114	Barbeque	1
115	Chipotle	1
116	Smoked	1
117	Mexican	1
118	Btroot	1
119	Fries	1

	WORD	COUNT
120	Roast	1
121	Dorito	1
122	Hony	1
123	OnionDip	1
124	Mexicana	1
125	Truffle	1
126	Sp	1
127	salt	1
128	Originl	1
129	Mzzrlla	1
130	FriedChicken	1
131	Sthrn	1
132	Chili	1
133	Gcamole	1
134	Crnchers	1
135	Crn	1
136	Infzns	1
137	Box	1
138	Southern	1
139	Garden	1
140	NCC	1
141	Siracha	1
142	Chp	1
143	Mozzarella	1
144	Tmato	1
145	Basil	1
146	Co	1
147	Tom	1
148	Chnky	1
149	Onion	1

	WORD	COUNT
150	SourCream	1
151	Pot	1
152	Jam	1
153	Jalapeno	1
154	Plus	1
155	GrnWves	1
156	Crackers	1
157	Prawn	1
158	Rib	1
159	Seasonedchicken	1
160	N	1
161	Mac	1
162	Fig	1
163	Camembert	1
164	PotatoMix	1
165	SweetChili	1
166	Pesto	1
167	Bolognese	1

```
In [121...] salsas = transactionData2[transactionData2['PROD_NAME'].str.contains("Salsa") == Tr
salsas['PROD_NAME'].unique()
# ALL need to be removed, other than Smith's Crinkle Cut Tomato Salsa 150g and Red
```

```
Out[121...] array(['Old El Paso Salsa Dip Tomato Mild 300g',
'Red Rock Deli SR Salsa & Mzzrlla 150g',
'Smiths Crinkle Cut Tomato Salsa 150g',
'Doritos Salsa Medium 300g',
'Old El Paso Salsa Dip Chnky Tom Ht300g',
'Woolworths Mild Salsa 300g',
'Old El Paso Salsa Dip Tomato Med 300g',
'Woolworths Medium Salsa 300g', 'Doritos Salsa Mild 300g'],
dtype=object)
```

```
In [122...] '''
transactionData2[transactionData2['PROD_NAME'].str.contains("Mac") == True]: Checke
transactionData2[transactionData2['PROD_NAME'].str.contains("RRD") == True]: Check,
transactionData2[transactionData2['PROD_NAME'].str.contains("Chicken") == True]: Ch
```


Note: Red Rock Deli and RDD are the same, need to check abbreviations

```
Check = transactionData2[transactionData2['PROD_NAME'].str.contains("Dip") == True]
Check['PROD_NAME'].unique()
```

Old El Paso Salsa Dip must be removed; Smith's Crinkle Cut French Onion Dip refers
Because I am noticing a trend with Old El Paso, I am going to isolate specified row

```
Old_El = transactionData2[transactionData2['PROD_NAME'].str.contains("Old") == True]
Old_El['PROD_NAME'].unique()
Old El Paso is only connected to Salsa
```

```
Doritos_salsa = transactionData2[transactionData2['PROD_NAME'].str.contains("Dorito") == True]
Doritos_salsa['PROD_NAME'].unique()
Doritos = transactionData2[transactionData2['PROD_NAME'].str.contains("Doritos") == True]
Doritos['PROD_NAME'].unique()
```

```
Woolworths = transactionData2[transactionData2['PROD_NAME'].str.contains("Woolworth") == True]
Woolworths['PROD_NAME'].unique()
```

Woolworths does not make potato chips; therefore can be removed

```
corn = transactionData2[transactionData2['PROD_NAME'].str.contains("Corn") == True]
corn['PROD_NAME'].unique()
```

All, including corn, do not relate to potato chips

```
WW = transactionData2[transactionData2['PROD_NAME'].str.contains("WW") == True]
WW['PROD_NAME'].unique()
```

WW is fine, removing Corn cleans WW rows

```
Strws = transactionData2[transactionData2['PROD_NAME'].str.contains("Strws") == True]
Strws['PROD_NAME'].unique()
```

```
Straws = transactionData2[transactionData2['PROD_NAME'].str.contains("Straws") == True]
Straws['PROD_NAME'].unique()
```

```
Tostitos = transactionData2[transactionData2['PROD_NAME'].str.contains("Tostitos") == True]
Tostitos['PROD_NAME'].unique()
```

```
Cheetos = transactionData2[transactionData2['PROD_NAME'].str.contains("Cheetos") == True]
Cheetos['PROD_NAME'].unique()
```

```
Bolognese = transactionData2[transactionData2['PROD_NAME'].str.contains("Bolognese") == True]
Bolognese['PROD_NAME'].unique()
'''
```

```
Keyword_removed = ['Old', 'Doritos', 'Whlgrn', 'Rings', 'Woolworths', 'Crackers', 'Corn', 'WW', 'Strws', 'Straws', 'Tostitos', 'Cheetos', 'Bolognese']
Cleaned_PC = transactionData2[transactionData2['PROD_NAME'].str.contains('|'.join(Keyword_removed)) == False]
Cleaned_PC['PROD_NAME'].value_counts()
```

```

Out[122...  PROD_NAME
Kettle Mozzarella Basil & Pesto 175g 3304
Cobs Popd Swt/Chilli &Sr/Cream Chips 110g 3269
Tyrrells Crisps Ched & Chives 165g 3268
Cobs Popd Sea Salt Chips 110g 3265
Kettle 135g Swt Pot Sea Salt 3257
Infuzions Thai SweetChili PotatoMix 110g 3242
Smiths Crinkle Chip Orgnl Big Bag 380g 3233
Thins Potato Chips Hot & Spicy 175g 3229
Kettle Sensations Camembert & Fig 150g 3219
Pringles Barbeque 134g 3210
Kettle Sweet Chilli And Sour Cream 175g 3200
Smiths Crinkle Chips Salt & Vinegar 330g 3197
Thins Chips Light& Tangy 175g 3188
Pringles Sweet&Spcy BBQ 134g 3177
Tyrrells Crisps Lightly Salted 165g 3174
Kettle Sea Salt And Vinegar 175g 3173
Pringles SourCream Onion 134g 3162
Cobs Popd Sour Crm &Chives Chips 110g 3159
Kettle Original 175g 3159
Pringles Original Crisps 134g 3157
Kettle Honey Soy Chicken 175g 3148
Infzns Crn Crnchers Tangy Gcamole 110g 3144
Smiths Crinkle Original 330g 3142
Kettle Sensations Siracha Lime 150g 3127
Pringles Mystery Flavour 134g 3114
Thins Chips Seasonedchicken 175g 3114
Pringles Chicken Salt Crips 134g 3104
Thins Chips Salt & Vinegar 175g 3103
Pringles Slt Vingar 134g 3095
Kettle Sensations BBQ&Maple 150g 3083
Pringles Sthrn FriedChicken 134g 3083
Kettle Chilli 175g 3038
Smiths Chip Thinly Cut Original 175g 1614
Natural Chip Co Tmato Hrb&Spce 175g 1572
Natural ChipCo Sea Salt & Vinegr 175g 1550
RRD SR Slow Rst Pork Belly 150g 1526
Smiths Thinly Cut Roast Chicken 175g 1519
RRD Sweet Chilli & Sour Cream 165g 1516
RRD Honey Soy Chicken 165g 1513
Smith Crinkle Cut Mac N Cheese 150g 1512
Infuzions Mango Chutny Papadums 70g 1507
RRD Chilli& Coconut 150g 1506
Smiths Crinkle Cut Snag&Sauce 150g 1503
Red Rock Deli Sp Salt & Truffle 150G 1498
Red Rock Deli Thai Chilli&Lime 150g 1495
Smiths Crinkle Cut Chips Barbecue 170g 1489
WW Original Stacked Chips 160g 1487
Smiths Crinkle Cut Chips Chicken 170g 1484
WW Sour Cream &OnionStacked Chips 160g 1483
Smiths Crinkle Cut Chips Chs&Onion170g 1481
RRD Salt & Vinegar 165g 1474
RRD Lime & Pepper 165g 1473
Smiths Chip Thinly S/Cream&Onion 175g 1473
Smiths Crinkle Cut Tomato Salsa 150g 1470
WW D/Style Chip Sea Salt 200g 1469

```

```

Natural Chip      Compny SeaSalt175g      1468
WW Crinkle Cut   Chicken 175g              1467
Smiths Crinkle Cut Chips Original 170g      1461
Smiths Thinly    Swt Chli&S/Cream175G      1461
Natural ChipCo   Hony Soy Chckn175g        1460
Red Rock Deli SR Salsa & Mzzrlla 150g      1458
Smiths Crinkle Cut Salt & Vinegar 170g      1455
RRD Steak &      Chimuchurri 150g          1455
Smith Crinkle Cut Bolognese 150g           1451
Thins Chips      Originl saltd 175g        1441
Smiths Chip Thinly CutSalt/Vinegr175g      1440
Smiths Crinkle Cut French OnionDip 150g      1438
Red Rock Deli Chikn&Garlic Aioli 150g      1434
RRD Pc Sea Salt  165g                    1431
NCC Sour Cream & Garden Chives 175g        1419
French Fries Potato Chips 175g              1418
WW Crinkle Cut   Original 175g             1410
Name: count, dtype: int64

```

In [123... `Cleaned_PC.head()`

Out[123...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_I
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
2	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
6	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
10	2019-05-17	7	7215	7176	16	Smiths Crinkle Chips Salt & Vinegar 330g	

In [124... `Cleaned_PC.describe()`

Out [124...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR
count	160718	160718.000000	1.607180e+05	1.607180e+05	160718.000000
mean	2018-12-30 04:02:55.611941376	135.033182	1.354815e+05	1.351106e+05	55.227840
min	2018-07-01 00:00:00	1.000000	1.000000e+03	1.000000e+00	1.000000
25%	2018-09-30 00:00:00	70.000000	7.005200e+04	6.780425e+04	26.000000
50%	2018-12-30 00:00:00	130.000000	1.302640e+05	1.347405e+05	48.000000
75%	2019-03-31 00:00:00	203.000000	2.030390e+05	2.024408e+05	85.000000
max	2019-06-30 00:00:00	272.000000	2.373711e+06	2.415841e+06	114.000000
std	NaN	76.671100	8.014557e+04	7.809281e+04	34.558011

In [125...

```
# Filter Data to find outliers
# Produce QTY verification: numbers should be at the lowest 1
print(f"MinProductQuantity: {Cleaned_PC['PROD_QTY'].min()} \nMaxProductQuantity: {C
MinProductQuantity: 1
MaxProductQuantity: 5
```

In [126...

```
# Produce QTY verification: numbers should be at the lowest 1
print(f"MinProductQuantity: {transactionData2['PROD_QTY'].min()} \nMaxProductQuanti
MinProductQuantity: 1
MaxProductQuantity: 200
```

In [127...

```
transactionData2[ transactionData2['PROD_QTY'] == 200 ]
```

Out [127...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRC
69762	2018-08-19	226	226000	226201	4	Dorito Corn Chp Supreme 380g	
69763	2019-05-20	226	226000	226210	4	Dorito Corn Chp Supreme 380g	

In [128...

```
'''
The maximum we seen ealier were corn chips not potato, so it was removed from analy
Based on the purchase data and these two being the same amount, this might be for c
'''
```

Out[128... 'The maximum we seen earlier were corn chips not potato, so it was removed from a
analysis, however this seemed to be the only outlier. Based on the purchase data a
nd these two being the same amount, this might be for commercial use.'

```
In [129... Cleaned_PC.nunique()

print(f"""
Cleaned Data Summary:
-----
Time Period    : 1 year of data
Products       : {Cleaned_PC['PROD_NAME'].nunique()} unique chip flavors/brands
Stores         : {Cleaned_PC['STORE_NBR'].nunique()} unique stores
Customers      : {Cleaned_PC['LYLTY_CARD_NBR'].nunique()} unique customers
Transactions   : {Cleaned_PC['TXN_ID'].nunique()} unique transactions
""")
```

```
Cleaned Data Summary:
-----
Time Period    : 1 year of data
Products       : 72 unique chip flavors/brands
Stores         : 269 unique stores
Customers      : 62005 unique customers
Transactions   : 160118 unique transactions
```

```
In [130... customerData.head()
```

```
Out[130... 
```

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
0	1000	YOUNG SINGLES/COUPLES	Premium
1	1002	YOUNG SINGLES/COUPLES	Mainstream
2	1003	YOUNG FAMILIES	Budget
3	1004	OLDER SINGLES/COUPLES	Mainstream
4	1005	MIDAGE SINGLES/COUPLES	Mainstream

```
In [131... customerData.describe()
```

Out[131...

LYLTY_CARD_NBR

count	7.263700e+04
mean	1.361859e+05
std	8.989293e+04
min	1.000000e+03
25%	6.620200e+04
50%	1.340400e+05
75%	2.033750e+05
max	2.373711e+06

In [132...

```
customerData['LIFESTAGE'].value_counts()
```

Out[132...

```
LIFESTAGE
RETIREES          14805
OLDER SINGLES/COUPLES  14609
YOUNG SINGLES/COUPLES  14441
OLDER FAMILIES      9780
YOUNG FAMILIES      9178
MIDAGE SINGLES/COUPLES  7275
NEW FAMILIES        2549
Name: count, dtype: int64
```

In [133...

```
customerData['PREMIUM_CUSTOMER'].value_counts()
```

Out[133...

```
PREMIUM_CUSTOMER
Mainstream    29245
Budget        24470
Premium       18922
Name: count, dtype: int64
```

In [134...

```
customerData.isnull().sum()
```

Out[134...

```
LYLTY_CARD_NBR    0
LIFESTAGE          0
PREMIUM_CUSTOMER  0
dtype: int64
```

In [135...

```
concat_data = Cleaned_PC.merge(customerData, on = 'LYLTY_CARD_NBR')
```

In [136...

```
concat_data.head()
```

Out[136...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_Q
0	2018-10-17	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	2019-05-20	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
2	2018-08-17	2	2373	974	69	Smiths Chip Thinly S/ Cream&Onion 175g	
3	2019-05-16	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
4	2019-05-17	7	7215	7176	16	Smiths Crinkle Chips Salt & Vinegar 330g	

In [137...

```
concat_data.isnull().sum()
```

Out[137...

```
DATE                0
STORE_NBR           0
LYLTY_CARD_NBR      0
TXN_ID              0
PROD_NBR            0
PROD_NAME           0
PROD_QTY            0
TOT_SALES           0
LIFESTAGE           0
PREMIUM_CUSTOMER    0
dtype: int64
```

In [138...

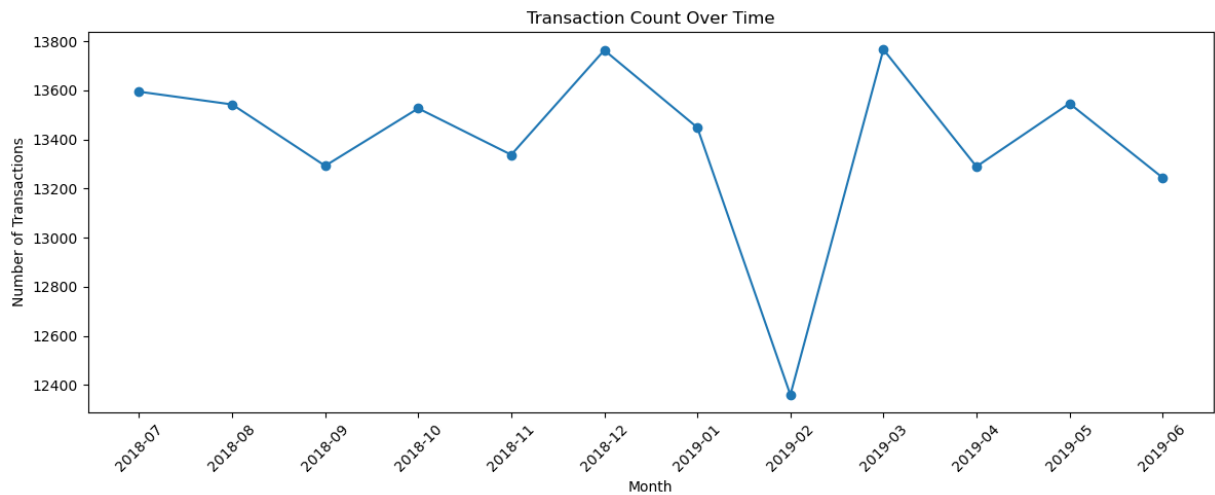
```
import matplotlib.pyplot as plt
```

In [139...

```
concat_data['DATE'] = pd.to_datetime(concat_data['DATE'])
concat_data = concat_data.sort_values('DATE')

monthly_counts = concat_data.groupby(concat_data['DATE'].dt.to_period('M')).size()
monthly_counts['DATE'] = monthly_counts['DATE'].astype(str)

plt.figure(figsize=(12, 5))
plt.plot(monthly_counts['DATE'], monthly_counts['count'], marker='o')
plt.title('Transaction Count Over Time')
plt.xlabel('Month')
plt.ylabel('Number of Transactions')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



PACK SIZE

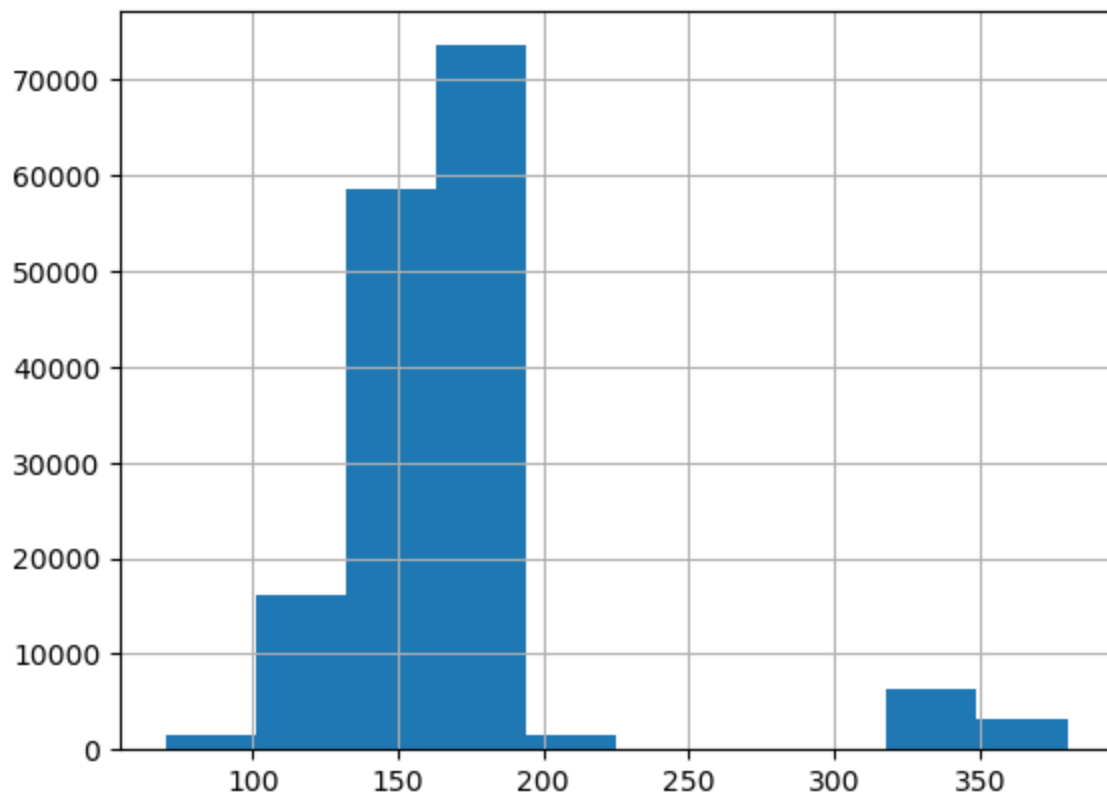
```
In [140...] concat_data['PACK_SIZE'] = concat_data['PROD_NAME'].str.extract(r'(\d+)').astype(int)
```

```
In [141...] print(f"Max pack size: {concat_data['PACK_SIZE'].max()} \nMin pack size: {concat_da
```

Max pack size: 380

Min pack size: 70

```
In [142...] concat_data['PACK_SIZE'].hist()  
plt.show()
```



BRANDS


```
In [143... concat_data['BRAND'] = concat_data['PROD_NAME'].str.split().str[0]
```

```
In [144... concat_data.head()
```

```
Out[144... 
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
72519	2018-07-01	15	15200	13928	109	Pringles Barbeque 134g	
20236	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	
132941	2018-07-01	43	43180	39840	26	Pringles Sweet&Spcy BBQ 134g	
120511	2018-07-01	186	186426	188794	36	Kettle Chilli 175g	
16852	2018-07-01	238	238384	243454	32	Kettle Sea Salt And Vinegar 175g	

```
In [145... concat_data['BRAND'].unique()
```

```
Out[145... array(['Pringles', 'Smith', 'Kettle', 'RRD', 'Tyrrells', 'WW', 'Smiths',  
      'Red', 'Infzns', 'French', 'Thins', 'Natural', 'Cobs', 'Infuzions',  
      'NCC'], dtype=object)
```

```
In [146... brand_mapping = {  
    'NCC'      : 'Nice & Natural',  
    'Thins'    : 'Thins',  
    'WW'       : 'Woolworths',  
    'Smiths'   : 'Smiths',  
    'Kettle'   : 'Kettle',  
    'French'   : 'French',  
    'Cobs'     : 'Cobs',  
    'Infuzions': 'Infuzions',  
    'RRD'      : 'Red Rock Deli',  
    'Pringles' : 'Pringles',  
    'Infzns'   : 'Infuzions',      # Likely a typo/duplicate  
    'Tyrrells' : 'Tyrrells',  
    'Smith'    : 'Smiths',         # Likely a typo/duplicate  
    'Red'      : 'Red Rock Deli',  # Likely part of Red Rock Deli  
    'Natural'  : 'Natural Chip Co'  
}  
  
# Apply it  
concat_data['BRAND'] = concat_data['BRAND'].map(brand_mapping)
```

```
In [147... concat_data.head()
```

Out[147...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
72519	2018-07-01	15	15200	13928	109	Pringles Barbeque 134g	
20236	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	
132941	2018-07-01	43	43180	39840	26	Pringles Sweet&Spcy BBQ 134g	
120511	2018-07-01	186	186426	188794	36	Kettle Chilli 175g	
16852	2018-07-01	238	238384	243454	32	Kettle Sea Salt And Vinegar 175g	

In [148...

```
concat_data['PRICE'] = concat_data['TOT_SALES'] / concat_data['PROD_QTY']  
concat_data.head()
```

Out[148...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
72519	2018-07-01	15	15200	13928	109	Pringles Barbeque 134g	
20236	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	
132941	2018-07-01	43	43180	39840	26	Pringles Sweet&Spcy BBQ 134g	
120511	2018-07-01	186	186426	188794	36	Kettle Chilli 175g	
16852	2018-07-01	238	238384	243454	32	Kettle Sea Salt And Vinegar 175g	

Distribution of customers by each group

In [149...

```
# Life Stage Ranked and Count of Customers in each group  
concat_data.groupby('LIFESTAGE').size().reset_index(name = 'count').sort_values('co  
# Older Singles/couples spend the most on potato chips
```

Out[149...

	LIFESTAGE	count
3	OLDER SINGLES/COUPLES	32914
4	RETIREEES	29996
2	OLDER FAMILIES	29947
5	YOUNG FAMILIES	26587
6	YOUNG SINGLES/COUPLES	21900
0	MIDAGE SINGLES/COUPLES	15207
1	NEW FAMILIES	4167

In [151...

```
# Premium Ranked and count of customers in each group
concat_data.groupby('PREMIUM_CUSTOMER').size().reset_index(name = 'count').sort_val
# Mainstream spends the most on potato chips
```

Out[151...

	PREMIUM_CUSTOMER	count
1	Mainstream	61434
0	Budget	56744
2	Premium	42540

In [152...

```
# Combined ranked, Premium, and Life Stage
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER']).size().reset_index(name = 'co
```

Out [152...

	LIFESTAGE	PREMIUM_CUSTOMER	count
6	OLDER FAMILIES	Budget	14267
13	RETIREEES	Mainstream	12919
19	YOUNG SINGLES/COUPLES	Mainstream	12300
15	YOUNG FAMILIES	Budget	11577
9	OLDER SINGLES/COUPLES	Budget	11144
10	OLDER SINGLES/COUPLES	Mainstream	11088
11	OLDER SINGLES/COUPLES	Premium	10682
12	RETIREEES	Budget	9145
7	OLDER FAMILIES	Mainstream	8742
14	RETIREEES	Premium	7932
16	YOUNG FAMILIES	Mainstream	7879
17	YOUNG FAMILIES	Premium	7131
1	MIDAGE SINGLES/COUPLES	Mainstream	7110
8	OLDER FAMILIES	Premium	6938
18	YOUNG SINGLES/COUPLES	Budget	5711
2	MIDAGE SINGLES/COUPLES	Premium	5023
20	YOUNG SINGLES/COUPLES	Premium	3889
0	MIDAGE SINGLES/COUPLES	Budget	3074
3	NEW FAMILIES	Budget	1826
4	NEW FAMILIES	Mainstream	1396
5	NEW FAMILIES	Premium	945

Distribution of Sales by each group

In [153...

```
# Combined Group
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['TOT_SALES'].sum().reset_ind
```

Out[153...

	LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES
6	OLDER FAMILIES	Budget	101083.8
13	RETIREEES	Mainstream	91124.7
19	YOUNG SINGLES/COUPLES	Mainstream	90177.1
15	YOUNG FAMILIES	Budget	82198.0
9	OLDER SINGLES/COUPLES	Budget	80796.4
10	OLDER SINGLES/COUPLES	Mainstream	78695.7
11	OLDER SINGLES/COUPLES	Premium	77341.9
12	RETIREEES	Budget	66196.6
7	OLDER FAMILIES	Mainstream	62002.6
14	RETIREEES	Premium	57657.2
16	YOUNG FAMILIES	Mainstream	55445.1
1	MIDAGE SINGLES/COUPLES	Mainstream	52883.5
17	YOUNG FAMILIES	Premium	50659.3
8	OLDER FAMILIES	Premium	48875.1
18	YOUNG SINGLES/COUPLES	Budget	36850.0
2	MIDAGE SINGLES/COUPLES	Premium	34934.2
20	YOUNG SINGLES/COUPLES	Premium	25316.2
0	MIDAGE SINGLES/COUPLES	Budget	21116.0
3	NEW FAMILIES	Budget	12972.8
4	NEW FAMILIES	Mainstream	9794.0
5	NEW FAMILIES	Premium	6745.3

In [154...

```
# Member Status Only
concat_data.groupby('PREMIUM_CUSTOMER')['TOT_SALES'].sum().reset_index().sort_value
```

Out[154...

	PREMIUM_CUSTOMER	TOT_SALES
1	Mainstream	440122.7
0	Budget	401213.6
2	Premium	301529.2

In [155...

```
# Lifestage only
concat_data.groupby('LIFESTAGE')['TOT_SALES'].sum().reset_index().sort_values('TOT_
```

Out[155...

	LIFESTAGE	TOT_SALES
3	OLDER SINGLES/COUPLES	236834.0
4	RETIREEES	214978.5
2	OLDER FAMILIES	211961.5
5	YOUNG FAMILIES	188302.4
6	YOUNG SINGLES/COUPLES	152343.3
0	MIDAGE SINGLES/COUPLES	108933.7
1	NEW FAMILIES	29512.1

Quantity of Chips

In [156...

```
# Combined Group
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['PROD_QTY'].sum().reset_inde
```

Out[156...

	LIFESTAGE	PREMIUM_CUSTOMER	PROD_QTY
6	OLDER FAMILIES	Budget	27751
13	RETIREEES	Mainstream	24332
19	YOUNG SINGLES/COUPLES	Mainstream	22760
15	YOUNG FAMILIES	Budget	22480
9	OLDER SINGLES/COUPLES	Budget	21332
10	OLDER SINGLES/COUPLES	Mainstream	21168
11	OLDER SINGLES/COUPLES	Premium	20430
12	RETIREEES	Budget	17292
7	OLDER FAMILIES	Mainstream	17030
16	YOUNG FAMILIES	Mainstream	15296
14	RETIREEES	Premium	15088
17	YOUNG FAMILIES	Premium	13806
1	MIDAGE SINGLES/COUPLES	Mainstream	13588
8	OLDER FAMILIES	Premium	13504
18	YOUNG SINGLES/COUPLES	Budget	10259
2	MIDAGE SINGLES/COUPLES	Premium	9491
20	YOUNG SINGLES/COUPLES	Premium	6997
0	MIDAGE SINGLES/COUPLES	Budget	5787
3	NEW FAMILIES	Budget	3388
4	NEW FAMILIES	Mainstream	2572
5	NEW FAMILIES	Premium	1757

In [158...

```
# Member Type
concat_data.groupby('PREMIUM_CUSTOMER')['PROD_QTY'].sum().reset_index().sort_values
```

Out[158...

	PREMIUM_CUSTOMER	PROD_QTY
1	Mainstream	116746
0	Budget	108289
2	Premium	81073

In [159...

```
# Lifestage
concat_data.groupby('LIFESTAGE')['PROD_QTY'].sum().reset_index().sort_values('PROD_
```

Out[159...

	LIFESTAGE	PROD_QTY
3	OLDER SINGLES/COUPLES	62930
2	OLDER FAMILIES	58285
4	RETIREEES	56712
5	YOUNG FAMILIES	51582
6	YOUNG SINGLES/COUPLES	40016
0	MIDAGE SINGLES/COUPLES	28866
1	NEW FAMILIES	7717

Average Chip Price

In [160...

```
# Combined
concat_data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['PRICE'].mean().reset_index()
```


Out[160...

	LIFESTAGE	PREMIUM_CUSTOMER	PRICE
19	YOUNG SINGLES/COUPLES	Mainstream	3.954472
1	MIDAGE SINGLES/COUPLES	Mainstream	3.890816
5	NEW FAMILIES	Premium	3.825926
3	NEW FAMILIES	Budget	3.824918
12	RETIREEES	Budget	3.819776
14	RETIREEES	Premium	3.816818
4	NEW FAMILIES	Mainstream	3.782808
9	OLDER SINGLES/COUPLES	Budget	3.781802
11	OLDER SINGLES/COUPLES	Premium	3.780481
13	RETIREEES	Mainstream	3.736984
10	OLDER SINGLES/COUPLES	Mainstream	3.712040
17	YOUNG FAMILIES	Premium	3.672063
2	MIDAGE SINGLES/COUPLES	Premium	3.670456
15	YOUNG FAMILIES	Budget	3.656165
7	OLDER FAMILIES	Mainstream	3.642610
6	OLDER FAMILIES	Budget	3.641971
0	MIDAGE SINGLES/COUPLES	Budget	3.641786
16	YOUNG FAMILIES	Mainstream	3.626507
8	OLDER FAMILIES	Premium	3.618867
20	YOUNG SINGLES/COUPLES	Premium	3.593597
18	YOUNG SINGLES/COUPLES	Budget	3.566731

In [161...

```
# membership
concat_data.groupby('PREMIUM_CUSTOMER')['PRICE'].mean().reset_index().sort_values('
```

Out[161...

	PREMIUM_CUSTOMER	PRICE
1	Mainstream	3.767273
2	Premium	3.713657
0	Budget	3.699288

In [162...

```
# Lifestage
concat_data.groupby('LIFESTAGE')['PRICE'].mean().reset_index().sort_values('PRICE',
```

Out[162...

	LIFESTAGE	PRICE
1	NEW FAMILIES	3.811039
6	YOUNG SINGLES/COUPLES	3.789274
4	RETIREEES	3.783336
0	MIDAGE SINGLES/COUPLES	3.767689
3	OLDER SINGLES/COUPLES	3.757872
5	YOUNG FAMILIES	3.651640
2	OLDER FAMILIES	3.636805

Mainstream midage and young singles and couples are more willing to pay more per packet of chips compared to their budget and premium counterparts. This may be due to premium shoppers being more likely to buy healthy snacks and when they buy chips, this is mainly for entertainment purposes rather than their own consumption. This is also supported by there being fewer premium midage and young singles and couples buying chips compared to their mainstream counterparts.

In [163...

```
concat_data[concat_data['PREMIUM_CUSTOMER'] == 'Premium'].groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])
```

Out[163...

	LIFESTAGE	PREMIUM_CUSTOMER	count
3	OLDER SINGLES/COUPLES	Premium	10682
4	RETIREEES	Premium	7932
5	YOUNG FAMILIES	Premium	7131
2	OLDER FAMILIES	Premium	6938
0	MIDAGE SINGLES/COUPLES	Premium	5023
6	YOUNG SINGLES/COUPLES	Premium	3889
1	NEW FAMILIES	Premium	945

As the difference in average price per unit isn't large, we can check if this difference is statistically different.

In [164...

```
from scipy import stats

# Filter groups
mainstream_midage = concat_data[
    (concat_data['PREMIUM_CUSTOMER'] == 'Mainstream') &
    (concat_data['LIFESTAGE'] == 'MIDAGE SINGLES/COUPLES')
]['PRICE']

budget_premium_midage = concat_data[
    (concat_data['PREMIUM_CUSTOMER'].isin(['Premium', 'Budget'])) &
```

```

    (concat_data['LIFESTAGE'] == 'MIDAGE SINGLES/COUPLES')
][['PRICE']]

# T-test for Midage Singles/Couples
t_stat, p_value = stats.ttest_ind(mainstream_midage, budget_premium_midage)
print(f"Midage Singles/Couples - t-stat: {t_stat:.4f}, p-value: {p_value:.4f}")

# Filter groups
mainstream_young = concat_data[
    (concat_data['PREMIUM_CUSTOMER'] == 'Mainstream') &
    (concat_data['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES')
][['PRICE']]

budget_premium_young = concat_data[
    (concat_data['PREMIUM_CUSTOMER'].isin(['Premium', 'Budget'])) &
    (concat_data['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES')
][['PRICE']]

# T-test for Young Singles/Couples
t_stat, p_value = stats.ttest_ind(mainstream_young, budget_premium_young)
print(f"Young Singles/Couples - t-stat: {t_stat:.4f}, p-value: {p_value:.4f}")

```

Midage Singles/Couples - t-stat: 13.8078, p-value: 0.0000

Young Singles/Couples - t-stat: 27.3317, p-value: 0.0000

```

In [165... print("=== Midage Singles/Couples ===")
print(f"Mainstream mean price:      {mainstream_midage.mean():.4f}")
print(f"Budget/Premium mean price: {budget_premium_midage.mean():.4f}")

print("\n=== Young Singles/Couples ===")
print(f"Mainstream mean price:      {mainstream_young.mean():.4f}")
print(f"Budget/Premium mean price: {budget_premium_young.mean():.4f}")

```

```

=== Midage Singles/Couples ===
Mainstream mean price:      3.8908
Budget/Premium mean price:  3.6596

```

```

=== Young Singles/Couples ===
Mainstream mean price:      3.9545
Budget/Premium mean price:  3.5776

```

The t-test results in a p-value < 0.01, i.e., the unit price for mainstream young and mid-aged singles and couples is significantly higher than that of budget or premium, young and middle-aged singles and couples.

Deep Drive into segments: Older Families Budget

```

In [166... segment_1 = concat_data[(concat_data['PREMIUM_CUSTOMER'] == 'Budget') & (concat_data[
segment_1.head()

```

Out[166...

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRC
20236	2018-07-01	10	10111	9693	82	Smith Crinkle Cut Mac N Cheese 150g	
31173	2018-07-01	219	219154	218894	43	Smith Crinkle Cut Bolognese 150g	
31372	2018-07-01	223	223082	223402	64	Red Rock Deli SR Salsa & Mzzrlla 150g	
30616	2018-07-01	209	209073	207890	96	WW Original Stacked Chips 160g	
20038	2018-07-01	5	5085	4863	11	RRD Pc Sea Salt 165g	

In [167...

```
# Which brand is most purchased in the most transactions  
segment_1.groupby('BRAND').size().reset_index(name = 'count').sort_values('count',a
```

Out[167...

	BRAND	count
8	Smiths	3093
3	Kettle	2563
6	Pringles	1996
7	Red Rock Deli	1857
9	Thins	1171
11	Woolworths	773
0	Cobs	760
2	Infuzions	682
4	Natural Chip Co	576
10	Tyrrells	489
5	Nice & Natural	165
1	French	142

In [168...

```
# Which brand sold the most units  
segment_1.groupby('BRAND')['PROD_QTY'].sum().reset_index().sort_values('PROD_QTY',
```

Out[168...

	BRAND	PROD_QTY
8	Smiths	5999
3	Kettle	5008
6	Pringles	3865
7	Red Rock Deli	3604
9	Thins	2278
11	Woolworths	1510
0	Cobs	1489
2	Infuzions	1328
4	Natural Chip Co	1122
10	Tyrrells	953
5	Nice & Natural	319
1	French	276

In [169...

```
# Which brand is the most profitable for this segment  
segment_1.groupby('BRAND')['TOT_SALES'].sum().reset_index().sort_values('TOT_SALES'
```

Out[169...

	BRAND	TOT_SALES
3	Kettle	25231.6
8	Smiths	21651.1
6	Pringles	14300.5
7	Red Rock Deli	10162.2
9	Thins	7517.4
0	Cobs	5658.2
2	Infuzions	4660.0
10	Tyrrells	4002.6
4	Natural Chip Co	3366.0
11	Woolworths	2749.2
5	Nice & Natural	957.0
1	French	828.0

In [170...

```
segment_1.groupby('BRAND')['PRICE'].mean().reset_index().sort_values('PRICE', ascen
```

Out[170...

	BRAND	PRICE
3	Kettle	5.037924
10	Tyrrells	4.200000
0	Cobs	3.800000
6	Pringles	3.700000
8	Smiths	3.611219
2	Infuzions	3.508504
9	Thins	3.300000
1	French	3.000000
4	Natural Chip Co	3.000000
5	Nice & Natural	3.000000
7	Red Rock Deli	2.820194
11	Woolworths	1.821604

In [171...

```
segment_1.groupby('PACK_SIZE')['TOT_SALES'].sum().reset_index().sort_values('TOT_SA
```

Out[171...

	PACK_SIZE	TOT_SALES
8	175	34346.5
4	150	16403.8
2	134	14300.5
1	110	9655.8
6	165	8316.6
10	330	6150.3
7	170	4312.3
11	380	2991.3
3	135	2213.4
5	160	1128.6
0	70	662.4
9	200	602.3

Older Families Budget Insight: Although Smith's sold the most to Budget Older Families, Kettle generated the most revenue. Despite lower sales volume, Kettle's higher revenue suggests stronger brand value per unit, indicating a customer segment willing to pay a premium. This higher price point, however, appears to suppress overall demand. A controlled

price promotion on Kettle would allow the business to measure price elasticity and determine whether reducing price drives enough volume to offset the margin reduction, ultimately helping to identify whether price or preference is the primary driver of Kettle's lower unit sales. Furthermore, 175-gram size bags generate the most profit.

Deep dive into segments: Mainstream, young singles/couples

```
In [93]: segment_2 = concat_data[(concat_data['PREMIUM_CUSTOMER'] == 'Mainstream')&(concat_d
segment_2.head()
```

```
Out[93]:
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PR
154131	2018-07-01	212	212019	210612	107	Smiths Crinkle Cut French OnionDip 150g	
145361	2018-07-01	7	7200	7108	33	Cobs Popd Swt/Chlli &Sr/Cream Chips 110g	
156792	2018-07-01	272	272151	269848	26	Pringles Sweet&Spcy BBQ 134g	
146375	2018-07-01	29	29169	26180	28	Thins Potato Chips Hot & Spicy 175g	
155741	2018-07-01	245	245339	247845	7	Smiths Crinkle Original 330g	

```
In [94]: # Which brand is purchased during the most transactions
segment_2.groupby('BRAND').size().reset_index(name = 'count').sort_values('count',a
```

```
Out[94]:
```

	BRAND	count
1	Kettle	2949
2	Pringles	2315
3	Smiths	1988
4	Thins	1166
0	Infuzions	657
5	Tyrrells	619

```
In [95]: # Which brand sold the most units
```

```
segment_2.groupby('BRAND')['PROD_QTY'].sum().reset_index().sort_values('PROD_QTY',
```

Out[95]:

	BRAND	PROD_QTY
--	-------	----------

1	Kettle	5497
2	Pringles	4326
3	Smiths	3609
4	Thins	2187
0	Infuzions	1227
5	Tyrrells	1143

In [96]: *# Which brand is the most profitable for this segment*

```
segment_2.groupby('BRAND')['TOT_SALES'].sum().reset_index().sort_values('TOT_SALES'
```

Out[96]:

	BRAND	TOT_SALES
--	-------	-----------

1	Kettle	27718.6
2	Pringles	16006.2
3	Smiths	15265.7
4	Thins	7217.1
5	Tyrrells	4800.6
0	Infuzions	4508.6

In [98]:

```
segment_2.groupby('BRAND')['PRICE'].mean().reset_index().sort_values('PRICE', ascen
```

Out[98]:

	BRAND	PRICE
--	-------	-------

1	Kettle	5.042455
3	Smiths	4.205936
5	Tyrrells	4.200000
2	Pringles	3.700000
0	Infuzions	3.665753
4	Thins	3.300000

In [99]:

```
segment_2.groupby('PACK_SIZE')['TOT_SALES'].sum().reset_index().sort_values('TOT_SA
```


Out[99]:

	PACK_SIZE	TOT_SALES
8	175	29879.4
2	134	16006.2
4	150	11912.9
1	110	10389.2
6	165	7050.6
10	330	6378.3
11	380	3516.4
3	135	2247.0
7	170	1896.6
5	160	440.8
0	70	264.0
9	200	195.7

Mainstream Young Singles/Couples Insight: Kettle led across all key metrics; unit sales, total revenue. Price was consistent with findings across other segments. Unlike Budget Older Families, who showed sensitivity to Kettle's premium pricing, Mainstream Young Singles/Couples appear less price sensitive, willing to pay a premium for Kettle despite cheaper alternatives like Smiths being available. This is further supported by Kettle generating more revenue from Mainstream Young Singles/Couples than from Budget Older Families, suggesting that disposable income and lifestyle play a significant role in purchasing decisions. Notably, Pringles also ranked above Smiths in this segment, reinforcing the pattern that Mainstream Young Singles/Couples gravitate toward premium or well-known brands over value options. Pack size remained consistent with both segments.

In [172...

```
concat_data.to_csv('ChipDataCleaned.csv', index=False, encoding='utf-8-sig', sep=',
```

In []: